

6406533822514. ✓ I have written answers on the answer sheets

6406533822515. ✖ Not applicable

LLM

Section Id :	64065379954
Section Number :	7
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	12
Number of Questions to be attempted :	12
Section Marks :	40
Display Number Panel :	Yes
Section Negative Marks :	0
Group All Questions :	No
Enable Mark as Answered Mark for Review and Clear Response :	No
Section Maximum Duration :	0
Section Minimum Duration :	0
Section Time In :	Minutes
Maximum Instruction Time :	0
Sub-Section Number :	1
Sub-Section Id :	640653172026
Question Shuffling Allowed :	No

Question Number : 102 Question Id : 6406531130555 Question Type : MCQ

Correct Marks : 0

Question Label : Multiple Choice Question

THIS IS QUESTION PAPER FOR THE SUBJECT "DEGREE LEVEL : LARGE LANGUAGE MODELS (COMPUTER BASED EXAM)"

ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?

CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN.

(IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)

Options :

6406533822516. ✓ YES

6406533822517. ✖ NO

Question Number : 103 Question Id : 6406531130556 Question Type : MCQ

Correct Marks : 0

Question Label : Multiple Choice Question

General Note:

- If you find the given information is insufficient in any of the Numerical Answer Type (NAT) questions, enter -1 as the answer.
- In all your calculations, take at least two digits. For example, if the intermediate result is 1.860875, then take it as 1.86 to the next step.

Options :

6406533822518. ✓ Instructions has been mentioned above.

6406533822519. ✗ This Instructions is just for a reference & not for an evaluation.

Sub-Section Number :

2

Sub-Section Id :

640653172027

Question Shuffling Allowed :

Yes

Question Number : 104 Question Id : 6406531130557 Question Type : MCQ

Correct Marks : 2

Question Label : Multiple Choice Question

Let $R_E \in \mathbb{R}^{T_1 \times d_{model}}$ be the representation from the topmost encoder layer of the encoder and $Z_D \in \mathbb{R}^{T_2 \times d_{model}}$ be the output of Multihead Masked Attention layer of one of the decoder layers. What will be the size of the attention matrix in the Multi-head Cross Attention layer of the same decoder layer? Assume $d_q = d_v = d_k = d$ if required.

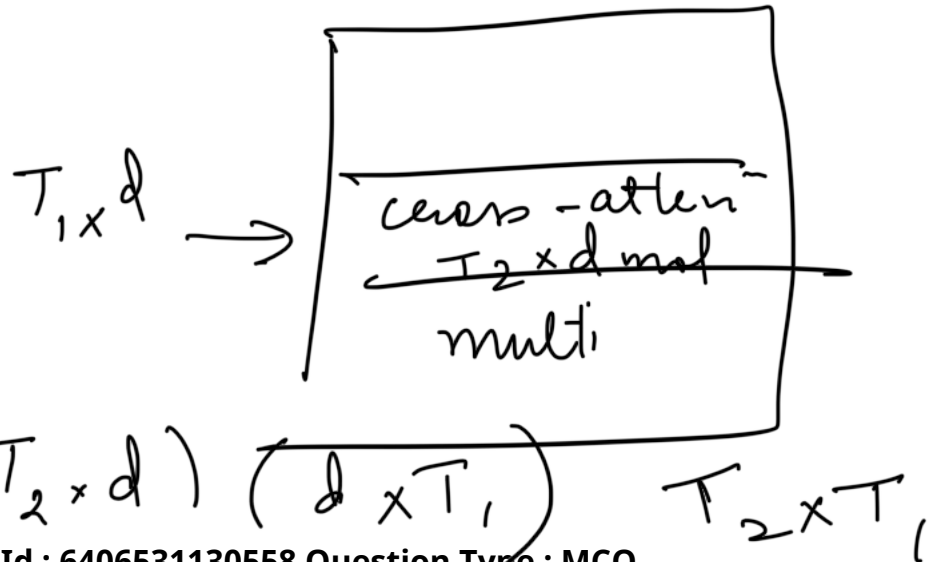
Options :

6406533822520. ✗ $T_1 \times T_1$

6406533822521. ✗ $T_1 \times T_2$

6406533822522. ✗ $T_2 \times T_2$

6406533822523. ✓ $T_2 \times T_1$



Question Number : 105 Question Id : 6406531130558 Question Type : MCQ

Correct Marks : 2

Question Label : Multiple Choice Question

Consider following assertion and reason pair:

Assertion: A transformer model cannot be trained in autoregressive mode for machine translation task.

Reason: Teacher forcing improves the convergence rate.

Choose the correct statements

Options :

6406533822524. ✖ Assertion and Reason are both true and Reason is a correct explanation of Assertion.
6406533822525. ✖ Assertion and Reason are both true and Reason is not a correct explanation of Assertion.
6406533822526. ✖ Assertion is true but Reason is false.
6406533822527. ✔ Assertion is false but Reason is true.

Sub-Section Number :

3

Sub-Section Id :

640653172028

Question Shuffling Allowed :

Yes

Question Number : 106 Question Id : 6406531130560 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

Suppose we use a pre-trained model for text generation with the given prompt "I am going to". Which of the following decoding strategies can be used such that the pre-trained model generates different text completion each time it is executed

Options :

6406533822532. ✖ Beam search with beam size 1
6406533822533. ✖ Greedy approach
6406533822534. ✔ Top-K with $k = 2$
6406533822535. ✖ None of these

Sub-Section Number :

4

Sub-Section Id :

640653172029

Question Shuffling Allowed :

Yes

Question Number : 107 Question Id : 6406531130561 Question Type : MSQ

Correct Marks : 2 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which of the following models, in general, struggle to encode a context when translating long sentences?

Options :

6406533822536. ✔ An RNN model
6406533822537. ✖ An RNN model with attention mechanism ~
6406533822538. ✖ A transformer model ~

Question Number : 108 Question Id : 6406531130562 Question Type : MSQ

Correct Marks : 2 Max. Selectable Options : 0

Question Label : Multiple Select Question

Choose the correct statements

Options :

6406533822539. ✓ All the parameters of the GPT model are randomly initialized during pre-training

6406533822540. ✗ The model minimizes the CLM objective during fine-tuning to improve the performance

6406533822541. ✗ All parameters of the GPT model are randomly initialized during fine-tuning

6406533822542. ✓ In general, fine-tuning requires a dataset with labels

Sub-Section Number :

Sub-Section Id :

Question Shuffling Allowed :

supervised
instruct - based fine-tuning
5
640653172030
Yes

Question Number : 109 Question Id : 6406531130559 Question Type : MSQ

Correct Marks : 3 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which of the following are correct statements regarding exhaustive decoding strategy? Note: $\mathcal{V}$ represents the vocabulary.

Options :

6406533822528. ✓ To generate a 10 token output, $|\mathcal{V}|^{10}$ possibilities will be considered.

6406533822529. ✗ To generate a 10 token output, $|\mathcal{V}|^{12}$ possibilities will be considered.

6406533822530. ✓ It is a deterministic approach.

6406533822531. ✗ It is a stochastic approach

degeneration
top-k

Question Number : 110 Question Id : 6406531130563 Question Type : MSQ

Correct Marks : 3 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which of the following decoding strategies is (are) appropriate for machine translation tasks? Assume we have infinite compute and the values for $k \geq 3, p > 0.35$ where required.

Options :

6406533822543. ✗ Top-k sampling

6406533822544. ✓ Beam search with the beam size $K = 2$

6406533822545. ✗ Top-p sampling

6406533822546. ✓ Exhaustive search

6406533822547. ✗ None of these

deterministic
approaches
work well on
compute

these things can
also work

Sub-Section Number :

6

Sub-Section Id :

640653172031

Question Shuffling Allowed :

No

Question Id : 6406531130564 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (111 to 113)

Question Label : Comprehension

Consider the following configuration for the Vannila transformer architecture with one encoder layer and one decoder layer.

- Source vocabulary size = 100
- Target vocabulary size = 200
- maximum sequence length = 32 $\Rightarrow T$
- length of context window (T) for both encoder and decoder = 32 T
- number of heads $n_h = 4$
- $df = 4 * d_{model} \rightarrow$ hidden layer of FFNN
- $dq = dk = dv = \frac{d_{model}}{n_h} \Rightarrow \frac{32}{4} = 8$

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 111 Question Id : 6406531130565 Question Type : SA

Correct Marks : 2

Question Label : Short Answer Question

Suppose the number of learnable parameters in the source input-embedding layer is 3200, how many parameters are there in the positional embedding layer of the source language?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

1024

context len x embed dim

$$\begin{array}{r} 32 \\ 32 \\ \hline 64 \\ 96 \times \end{array}$$

$$1024$$

Question Number : 112 Question Id : 6406531130566 Question Type : SA

Correct Marks : 3

Question Label : Short Answer Question

How many parameters are there in the multi-head attention layer of the encoder (exclude the parameters in the W_0 matrix used for linear transformation and FFN layer)?

Response Type : Numeric

$$d_{model} \times d_{k/q/v} \times W_k, W_q, W_v$$

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

3072

$$(32 \times 8) \times 3 \times 4 = 3072$$

↓ ↓
no. of heads

Question Number : 113 Question Id : 6406531130567 Question Type : SA

Correct Marks : 3

Question Label : Short Answer Question

How many parameters does the matrix W_Q have in anyone head of the encoder?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

256

$$32 \times 8 \rightarrow d_q$$

↓
256 → d_model

Sub-Section Number :

7

Sub-Section Id :

640653172032

Question Shuffling Allowed :

No

Question Id : 6406531130568 Question Type : COMPREHENSION Sub Question Shuffling

Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Numbers : (114 to 117)

Question Label : Comprehension

$$\mu = \frac{14}{4} = 3.5 \quad \sigma^2 = \frac{29}{4}$$

Consider following input matrix to a single neural network layer:

$$X = \begin{bmatrix} 1 & 2 & 3 & 8 \\ 4 & 7 & 11 & 14 \end{bmatrix}$$

$$2.5 = 6.25$$

$$1.5 = 2.25$$

$$0.5 = 0.25$$

$$4.5 = 20.25$$

It has 4 samples with 2 features each i.e. batch size is 4. Each column represents a sample. Left most column represents the first sample and topmost row represents the first feature vector.

Note: assume $\gamma = 1$, $\beta = 0$ and $\epsilon = 10^{-6}$ for both batch and layer normalization, if not mentioned explicitly. Use standard scaling.

$$\hat{x}_i = \frac{x_i - \mu_i}{\sqrt{\sigma^2 + \epsilon}}$$

Based on the above data, answer the given subquestions.

Sub questions

since, $\beta = 0, \gamma = 1$

Question Number : 114 Question Id : 6406531130569 Question Type : SA

Correct Marks : 1

$$y_i = x_i + \beta$$

Question Label : Short Answer Question

Say matrix F is computed by applying batch normalization on X , what will be sum of every element in first / topmost row of F ?

Response Type : Numeric

Evaluation Required For SA : Yes

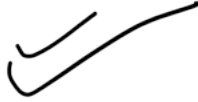
Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

0 to 0.01



$$\mu = 2.5$$
$$\sigma^2 = 2.5$$

$$\frac{x_i - \mu_i}{\sqrt{\sigma^2}}$$

$$\frac{-1.5}{\sqrt{2.5}} + \frac{1.5}{\sqrt{2.5}} = 0$$

Question Number : 115 Question Id : 6406531130570 Question Type : SA

Correct Marks : 1

Question Label : Short Answer Question

Say matrix F is computed by applying layer normalization on X , what will be sum of every element in first / leftmost column of F ?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

0 to 0.01



Question Number : 116 Question Id : 6406531130571 Question Type : SA

Correct Marks : 2

Question Label : Short Answer Question

Say σ_i 's were the standard deviations used to compute batch normalization of X . What is $\sum_{i=1} \sigma_i$?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

10.95 to 11.05

we calculate the

σ , in batch normal.

batch = 4 samples

if layer norm
calculate layer wise
(feature)

Question Number : 117 Question Id : 6406531130572 Question Type : SA

Correct Marks : 2

Question Label : Short Answer Question

Say σ_i 's were the standard deviations used to compute layer normalization of X .

What is $\sum_{i=1} \sigma_i$?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

6.45 to 6.55

6.5

Sub-Section Number :

8

Sub-Section Id :

640653172033

Question Shuffling Allowed :

No

Question Id : 6406531130573 **Question Type :** COMPREHENSION **Sub Question Shuffling Allowed :** No **Group Comprehension Questions :** No **Question Pattern Type :** NonMatrix

Question Numbers : (118 to 121)

Question Label : Comprehension

Consider following corpus (in italics), which is used train a GPT like (i.e. decoder only) model for causal language modeling:

Sachin Tendulkar is called God of Cricket. Sachin has broken many records. Sachin has highest number of centuries.

Sachin Pilgaonkar is a famous actor in Marathi cinema. Sachin has acted in, directed and produced numerous films. Sachin has broken many stereotypes.

Assume each word in the above corpus is a token. Tokens are case insensitive i.e. "The" is same as "the". Ignore special characters (e.g. period (.), semi colon (;), etc.)

Note: Estimate the joint probabilities using the Bayesian approach (no independence assumption) for the given corpus.

x_i represents i^{th} token from left to right. Tokens appear in italics.

Answer the given sub questions:

Sub questions

please note - for this

Question Number : 118 **Question Id :** 6406531130574 **Question Type :** SA **Calculate**

Correct Marks : 2

Question Label : Short Answer Question

total nos. of words in paragraph. And not total no. of tokens.

What is the value of $10 \times P(\text{Sachin})$?

Note $x_1 = \text{Sachin}$

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

1.37 to 1.47

$$P(s) = \frac{6}{42} \times 10 = 1.46$$



Question Number : 119 **Question Id :** 6406531130575 **Question Type :** SA

Correct Marks : 2

Question Label : Short Answer Question

What is the value of $100 \times P$
(*Sachin, Tendulkar*)?

Note $x_1 = \text{Sachin}$ and $x_2 = \text{Tendulkar}$

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

2.33 to 2.43

$$P(s, t) = \frac{2}{42} \times 100 = 2.38$$



Question Number : 120 **Question Id :** 6406531130576 **Question Type :** SA

Correct Marks : 3

Question Label : Short Answer Question

What is the value of the following
expression $100 \times P$
(*Sachin, has, broken, many, records*)

Note $x_1 = \text{Sachin}$, $x_2 = \text{has}$, and so on.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

2.33 to 2.43

$$P(s, h, b, m, r) = \frac{1}{42} \times 100 = 2.38$$

Question Number : 121 Question Id : 6406531130577 Question Type : MCQ

Correct Marks : 2

Question Label : Multiple Choice Question

Once the model is sufficiently trained, which of the following tokens will have the highest probability to be the next token, if the input is "Sachin has":

Options :

6406533822558. ✓ broken

6406533822559. ✗ acted

6406533822560. ✗ highest

6406533822561. ✗ None of these

since 'Sachin has
broken' comes twice
while 'Sachin has
acted' has come only
once

i-NLP

Section Id :	64065379955
Section Number :	8
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	31
Number of Questions to be attempted :	31
Section Marks :	50
Display Number Panel :	Yes
Section Negative Marks :	0
Group All Questions :	No
Enable Mark as Answered Mark for Review and Clear Response :	No
Section Maximum Duration :	0
Section Minimum Duration :	0
Section Time In :	Minutes
Maximum Instruction Time :	0
Sub-Section Number :	1
Sub-Section Id :	640653172034
Question Shuffling Allowed :	No

Question Number : 122 Question Id : 6406531130578 Question Type : MCQ

Correct Marks : 0

Question Label : Multiple Choice Question

THIS IS QUESTION PAPER FOR THE SUBJECT "DEGREE LEVEL : INTRODUCTION TO NATURAL LANGUAGE PROCESSING (COMPUTER BASED EXAM)"

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